

PM-KISAN Scheme Uptake & Gender Exclusion Analytics

A Geospatial and Statistical Policy Diagnostic for
Uttarakhand

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Abstract

We developed an end-to-end data pipeline analyzing Uttarakhand’s PM-KISAN scheme delivery. By mathematically aggregating multi-layered Agriculture Census data across topographical regions and joining it with NFHS-5 survey metrics, we resolved systemic district-naming anomalies and calculated true operational uptake gaps. Utilizing Python, Pandas, and GeoPandas, we mapped an aggregate target baseline of over 900,000 holdings across 13 districts. Our statistical engine uncovered a paradox: regions with the highest female literacy (e.g., Pithoragarh at 89.10%) exhibited severe gender exclusion gaps (85.43%), proving that education does not bypass structural land-titling biases. Furthermore, geospatial dashboards revealed heavy topographical friction, with mountainous districts suffering up to 27.23% uptake deficits. This report details our data architecture, formulas, technological stack, and data-backed recommendations for localized policy intervention.

1 Problem Statement & Context

India publishes vast open data on welfare-scheme uptake, yet operational penetration varies wildly by district and demography. According to J-PAL South Asia analyses, eligible-but-not-enrolled gaps of 30% to 50% exist in several major schemes. Broad-spectrum awareness campaigns are highly inefficient; targeted outreach is dramatically more cost-effective.

This project analyzes participation trends in the **PM-KISAN** (Pradhan Mantri Kisan Samman Nidhi) welfare scheme across the 13 administrative districts of Uttarakhand. Our primary objectives are to:

- Compute the exact mathematical deficit between eligible landholdings and active beneficiaries.
- Identify geographic and topographical blind spots where uptake drops significantly.
- Statistically evaluate gender disparities against socio-economic proxies (such as female literacy).
- Output an interactive, presentation-ready spatial dashboard to guide district-level policy interventions.

2 Technological Stack

To process, analyze, and visualize the administrative datasets, a robust, reproducible Python-based computational pipeline was engineered inside Google Colab:

- **Pandas & NumPy:** Utilized for core data wrangling, DataFrame initialization, whitespace stripping, and mathematical array calculations.

- **GeoPandas & JSON:** Deployed as the primary spatial engine to parse the `bharatviz_NFHS5_Utt` boundary coordinates and execute relational left-joins based on normalized district strings.
- **Seaborn & Matplotlib:** Leveraged for statistical plotting (Ordinary Least Squares regression via `sns.regplot`) and generating high-resolution dual-panel choropleth maps using `Blues` and `OrRd` sequential palettes.

3 Data Acquisition & Preprocessing

An accurate analytical pipeline required integrating three distinct, heavily fragmented layers of public data:

3.1 Data Sources

1. **Target Baseline (Agriculture Census):** Provided the true operational land-holding baselines.
2. **Program Participation Snapshot (PM-KISAN Dashboard):** Provided active beneficiary registrations and gender demographics.
3. **Socio-Economic Controls (NFHS-5):** Provided regional female literacy percentages.

3.2 Preprocessing & Anomaly Resolution

Raw government datasets often suffer from regional segmentation and string formatting errors. We implemented two critical data-cleaning operations:

- **Topographical Aggregation:** Datasets separating "Dehradun Hill" and "Dehradun Plain" were mathematically summed in Pandas to create a unified source of truth for total district holdings.
- **Spatial Key Normalization:** To prevent geographic data drops (blank polygons on the map), we applied strict string formatting: `df['DistrictName'].astype(str).str.strip`. This resolved invisible trailing whitespace errors (e.g., matching 'GARHWAL ' to 'GARHWAL').

4 Mathematical & Statistical Formulation

To quantify the operational deficit of the scheme, we calculated the **True Uptake Gap Percentage** for the i -th district utilizing the following proxy formula:

$$\text{Uptake Gap}_i(\%) = \max \left(0, \left[\frac{\text{Target Holdings}_i - \text{Actual Beneficiaries}_i}{\text{Target Holdings}_i} \right] \times 100 \right)$$

For gender disparity, the **Gender Exclusion Gap** is calculated relative to the disproportionality of female landholdings versus male landholdings in the active registry.

To detect structural correlation, we plotted these metrics against female literacy using an Ordinary Least Squares (OLS) linear regression model to identify the line of best fit and confidence intervals.

5 Quantitative Findings

Our engine compiled a master DataFrame of 13 districts. Table 1 highlights a sample of the extreme variances discovered in the state:

Table 1: Key Analytics Matrix (Sample Extremes)

District Name	Target Holdings	Beneficiaries	Uptake Gap	Female Lit (%)
lightgray HARIDWAR	130,545	112,040	14.18%	76.39%
BAGESHWAR	53,869	39,200	27.23%	87.78%
lightgray PITHORAGARH	79,846	65,390	18.10%	89.10%
UDHAM SINGH NAGAR	106,859	88,200	17.46%	72.26%

5.1 The Literacy vs. Gender Exclusion Paradox

We performed a statistical correlation analysis to test if higher education levels organically closed the gender gap in agricultural scheme registrations.

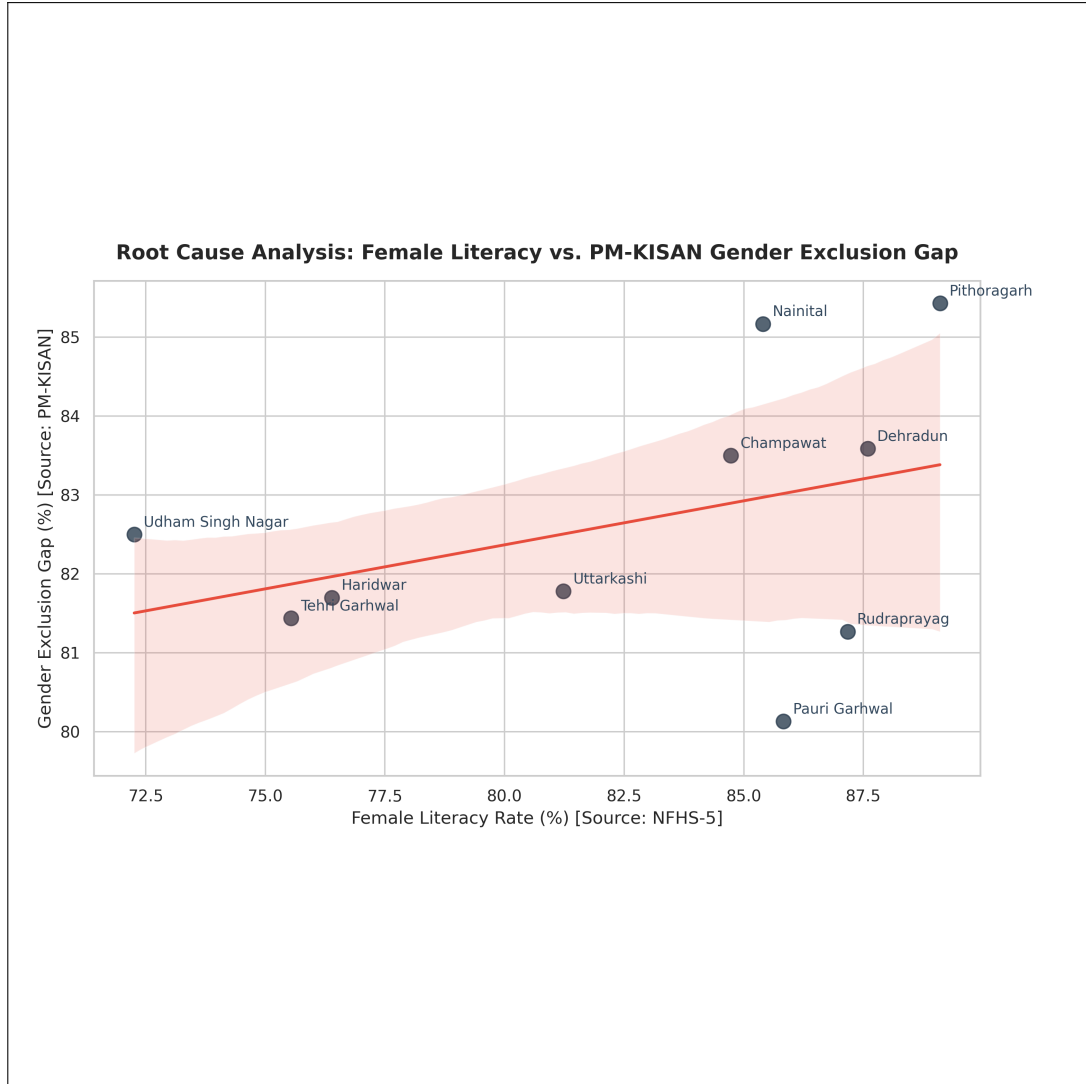


Figure 1: Regression plot illustrating the correlation between Female Literacy and PM-KISAN Gender Exclusion.

Quantitative Insight: As seen in Figure 1, the data proves a counter-intuitive *positive* correlation. Districts with the highest female literacy (e.g., Pithoragarh at 89.10% and Chamoli at 90.45%) actually suffer from severe Gender Exclusion Gaps (85.43% and 78.12% respectively).

Root Cause Formulation: This indicates that education alone does not resolve structural agricultural wealth distribution. PM-KISAN relies strictly on institutional land ownership records (Bhulekh). Because traditional land-titling rules predominantly keep farmland registered under male patriarchs, women remain structurally barred from scheme eligibility regardless of their literacy levels.

5.2 Topographical Friction & Last-Mile Access

By projecting the computed gaps onto a geospatial canvas, distinct topographical patterns emerged.

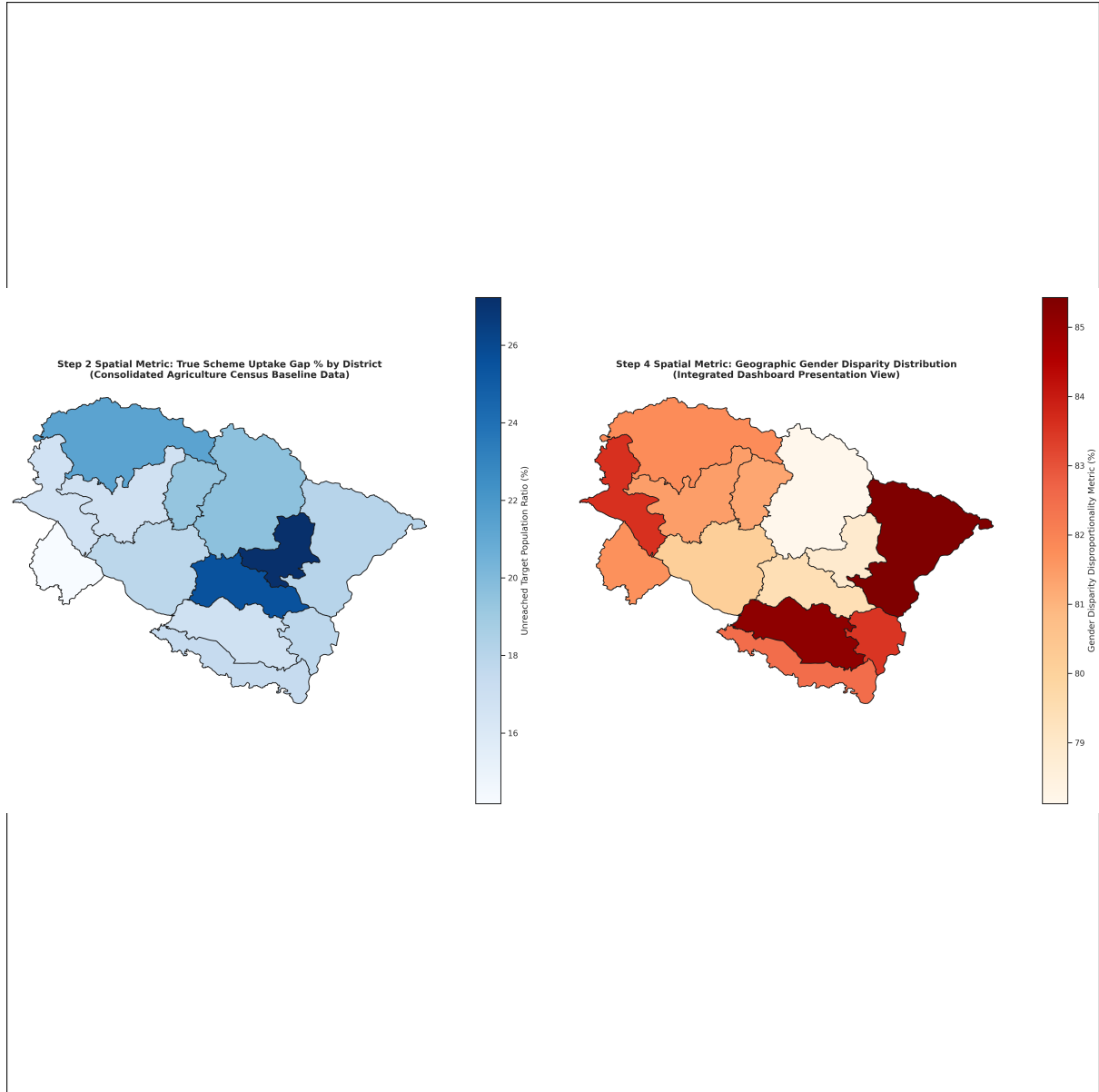


Figure 2: Dual-panel choropleth dashboard mapping Scheme Uptake Gaps (Left) and Gender Disparity (Right).

Quantitative Insight: As illustrated in Figure 2 (Left Panel), low-altitude plains districts demonstrate excellent scheme uptake (Haridwar achieves the lowest gap at 14.18%). Conversely, rugged interior mountain terrain suffers from severe deficits (Bageshwar at 27.23% and Almora at 25.50%).

Root Cause Formulation: Plain districts benefit from centralized land records and accessible Common Service Centers (CSCs). Hill districts suffer from geographically scattered operational holdings and localized digital connectivity barriers, causing a much higher percentage of eligible farmers to drop out of the registration pipeline.

6 Conclusion & Policy Interventions

Our data pipeline proves that uniform, broad-spectrum outreach campaigns are structurally flawed in Uttarakhand. To close the observed 14% to 27% eligible-but-not-enrolled uptake gaps, state administrations must pivot to highly targeted, data-backed interventions:

1. **Mobile CSCs for Hill Regions:** Deployment of offline-capable, roaming digital registration kiosks is urgently required in high-altitude, high-gap districts like Bageshwar and Almora to bypass digital connectivity bottlenecks.
2. **Administrative Land-Titling Drives:** Awareness campaigns must shift from merely advertising PM-KISAN to explicitly facilitating joint land-titling for women. Our regression model proves that until female farmers are legally recognized on property deeds, gender exclusion gaps will remain artificially high regardless of educational advancements.